## SEMESTER-IV

| UME405: COMPUTER AIDED DESIGN AND ANALYSIS | | | | |
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**Course Objective:** Introduce components and assemblies used in machines and use of 3D parametric CAD, CAE software for mechanical design. To provide an experiential learning environment using projects done by student groups, while applying CAD, CAE software tools to design mechanisms and structures for mechanical design evaluation, optimization of mass properties, static-stresses, deformations, etc. with experimental validation of simulation models.

**Syllabus**

**Standards, types, applications and working of following components and assemblies:**
**Machine Components:** Screw fasteners, Riveted joints, Keys, Cotters and joints, Shaft couplings, Pipe joints and fittings.

**Assemblies:** Bearings, Hangers and brackets, Steam and IC engine parts, Valves, Some important machine assemblies.

**Mechanical Drawing:** Machining and surface finish symbols and tolerances in dimensioning.

**CAD:** Introduction to CAD, CAM, CAE software in product life cycle.

**Geometric Modelling:** Parametric sketching and modelling, constrained model dimensioning, Relating dimensions and parameters. Feature and sequence of feature editing. Material addition and removal for extrude, revolve, blend, helical sweep, swept blend, variable section sweep. References and construction features of points, axis, curves, planes, surfaces. Cosmetic features, representation of welded joints, Draft and ribs features, chamfers, rounds, standard holes. Assembly modelling. Automatic production drawing creation and detailing for dimensions, BOM, Ballooning, sectioned views etc.

**Productivity Enhancement Tools in CAD Software:** Feature patterns, duplication, grouping, suppression. Top-down vs. bottom-up design.

**CAE:** Part and assembly mass property analysis. Customized analysis features. Sensitivity analysis of dimension and parameters, Automatic feasibility study and shape optimization.

**Mechanism Motion Analysis:** Kinematic joints used in mechanism assembly. Motion of kinematic chains, Plot coupler curve. Analysis of Mechanisms for interference, position, velocity, acceleration and bearing reactions.

**Analysis of Static Stress, Deflection, Temperature** etc. using software like 'Pro-Mechanica', 'SolidWorks Simulation'. Analysis of mechanical parts and assemblies. Using shells, beams and 2D for Plane strain/ plane stress or axisymmetric simplifications.
**Project:** Students will undertake projects individually or in groups to study the design of a simple mechanical system, make geometric models of the parts, assembly, evaluate the design and CAD automated drafting of production drawings of the system. The projects should be preferably based on experiential learning activities done. CAE analysis will be used to evaluate and redesign the system to optimize it for conditions of use. A technical report presenting and discussing the learnings from the project, will be the conclusion of the project. Projects could be mechanisms, simple machines / machine tools, simple products /

assemblies, structures studied in course of solids and structures / mechanics of machines, machine design etc.

**Course Learning Outcomes (CLOs)**

The students will be able to:

1. interpret mechanical drawings for components, assemblies and use parametric 3D CAD software tools in the correct manner for creating their geometric part models, assemblies and automated drawings.
2. create assembly of mechanism from schematic or component drawing and conduct position/ path/ kinematic / dynamic analysis of a mechanism in motion using CAD software tools.
3. evaluate design and create an optimized solution using commercial CAD, CAE software for required analysis of mass properties/ stress, deflection / temperature distribution etc. under realistic loading and constraining conditions.
4. produce design reports for geometric modelling, assembly, drawings, analysis, evaluation of results, reflection and suggestions for design evaluation and improvement

**Text Books**

1. Singh Ajeet, Machine Drawing, The McGraw-Hill Companies (2010)
2. Kelley David S., Pro/ENGINEER Wildfire #.0 Instructor, Tata McGraw Hill (2011), or of latest software release used in laboratory.
3. Shih Randy H., Introduction to Finite Element Analysis Using Creo Simulate #.0, SDC Publications, USA (ISBN: 978-1-58503-670-7, ISBN (Book + Software on Disk): 978- 1-58503-731-5 (2011), or of latest software release used in laboratory.

**Reference Books**

1. Gill, P.S., Machine Drawing, S.K.Kataria and Sons (2013).
2. Dhawan, R.K., Machine Drawing, S.Chand& Company Limited (2011).
3. Shyam Tikku and Prabhakar Singh, Pro/ENGINEER (Creo Parametric #.0) for Engineers and Designers, Dreamtech press (2013), or of latest software release used in laboratory.
4. Toogood Roger Ph.D., P. Eng., Zecher Jack P.E., Creo Parametric #.0 Tutorial and MultiMedia DVD, SDC Publications, USA (2012), ISBN: 978-1-58503-692-9, ISBN (Book + Software on Disk): 978-1-58503-730-8, or of latest software release used in laboratory.
5. Shih Randy H., Parametric Modeling with Creo Parametric #.0-An Introduction to Creo Parametric #.0, SDC Publications, USA (2011) ISBN: 978-1-58503-661-5, ISBN (Book + Software on Disk): 978-1-58503-729-2, or of latest software release used in laboratory.
6. Guide books in software help and online books at ptc.com

**Evaluation Scheme:**

| Sr. No. | Evaluation Elements | Weightage (%) |
|---|---|---|
| 1 | Sessional tests / assignments on software | 30 |
| 2 | Projects on modeling, assembly, drawing, Analysis of mass properties, stress, deflection, temperature, kinematics, dynamics etc. as relevant to the project. With technical reports of each. | 70 |

**NB:** 50% pass marks. Tests and projects on software will be open book examination.

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